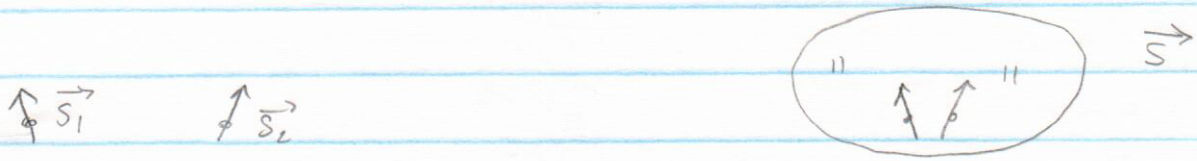
Angular Momentum Revisited



(I) 2 separate $S=\frac{1}{2}$ systems

(II) one object, a composite of 2 $S=\frac{1}{2}$ systems!

We will often want to consider the composite object (II) rather than the two $S=\frac{1}{2}$ systems separately. What can we say about the total angular momentum $\vec{S}=\vec{S}_1+\vec{S}_2$ quantum mechanically?

Example: Deuteron; nucleus = 1 proton + 1 neutron
($s=\frac{1}{2}$, $s=\frac{1}{2}$)

H_2 molecule — total molecular spin $(\vec{I}+\vec{S})$

Scattering: Total ang. mom. is conserved, but 3 components may flip; Example π-nucleon scattering

Simplest case Addition of Two spin-$\frac{1}{2}$ momenta

$\vec{S}_1$ = spin operator for particle 1
$\vec{S}_2$ = " " " " 2

Since 1, 2 are different particles, $[\vec{S}_1, \vec{S}_2]=0$

Pick quantization axis along $\vec{e}_z$, then we can use the eigenvectors of S_z in the uncoupled representation:

$|1\rangle = |\uparrow\rangle|\uparrow\rangle \equiv |\uparrow\uparrow\rangle$
$|2\rangle = |\uparrow\rangle|\downarrow\rangle \equiv |\uparrow\downarrow\rangle$
$|3\rangle = |\downarrow\rangle|\uparrow\rangle \equiv |\downarrow\uparrow\rangle$
$|4\rangle = |\downarrow\rangle|\downarrow\rangle \equiv |\downarrow\downarrow\rangle$

These constitute a basis for the coupled problem since they span the full space which is of dimension 4, and they are obviously linearly independent.

Let $\vec{S} = \vec{S}_1 + \vec{S}_2$. Note that

$$[S^i, S^j] = [S_1^i + S_2^i,\ S_1^j + S_2^j] = [S_1^i, S_1^j] + [S_2^i, S_2^j]$$

(cartesian component; which spin)

$$= i\hbar \epsilon_{ijk} [S_1^k + S_2^k]$$

$$= i\hbar \epsilon_{ijk} S^k$$

$\Rightarrow \vec{S}$ obeys the usual angular momentum commutation relations. Thus, on perfectly general grounds know that we can form simultaneous eigenstates of S^2, S_z, know that $S^2|\text{eigenstate}\rangle = s(s+1)|\text{eigenstate}\rangle$ etc!

Call the eigenvectors of S^2, S_z $|s, m\rangle$.

→ We want to figure out the $|s,m\rangle$ in terms of the uncoupled basis.

Dull way: Compute $\langle\mu|S^2|\nu\rangle$ & diagonalize

	$\|\uparrow\uparrow\rangle$	$\|\uparrow\downarrow\rangle$	$\|\downarrow\uparrow\rangle$	$\|\downarrow\downarrow\rangle$
$\langle\uparrow\uparrow\|$				
$\langle\uparrow\downarrow\|$				
$\langle\downarrow\uparrow\|$				
$\langle\downarrow\downarrow\|$				

Note that: $S_z|\uparrow\uparrow\rangle = (S_{1z}+S_{2z})|\uparrow\uparrow\rangle = (\frac{1}{2}+\frac{1}{2})|\uparrow\uparrow\rangle = |\uparrow\uparrow\rangle$

$S_z|\uparrow\downarrow\rangle = 0$

$S_z|\downarrow\uparrow\rangle = 0$

$S_z|\downarrow\downarrow\rangle = -|\downarrow\downarrow\rangle$

Eigenvalues of S_z are $0, \pm 1$. This means that $s=1, s=0$ (There is an $m=0$ "singlet" state with $s=0$ and an $s=1$ "triplet" state with $m=0,\pm1$: necessarily 3-fold degenerate)

We need a useful representation of S^2 so we can diag. it:

$$S^2 = (\vec{S_1}+\vec{S_2})^2 = S_1^2 + S_2^2 + 2\vec{S_1}\cdot\vec{S_2}$$

Raising & lowering ops

$$= \frac{3}{2} + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+} \qquad \left[S_{i\pm} = S_{ix} \pm iS_{iy}\right]$$

$S_-|sm\rangle = \sqrt{s(s+1)-m(m-1)}\,|s,m-1\rangle$ etc as usual.
$S_+|sm\rangle = \sqrt{s(s+1)-m(m+1)}\,|s,m+1\rangle$

$S_+|\uparrow\rangle = S_-|\downarrow\rangle = 0$

Note that $S^2|\uparrow\uparrow\rangle = \left(\frac{3}{2} + 2\cdot\frac{1}{2}\cdot\frac{1}{2}\right)|\uparrow\uparrow\rangle = 2|\uparrow\uparrow\rangle$

$S^2|\downarrow\downarrow\rangle = \left(\frac{3}{2} + 2\cdot\left(\frac{-1}{2}\right)\left(\frac{-1}{2}\right)\right)|\downarrow\downarrow\rangle = 2|\downarrow\downarrow\rangle$

→ $|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle$ are eigenstates of S^2 and $s=1$ for both.

$$S^2 = \frac{3}{2}\hat{1} + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+}$$

Note however that

$$S^2|\uparrow\downarrow\rangle = \left\{\frac{3}{2}|\uparrow\downarrow\rangle + 2\left(\tfrac{1}{2}\right)\left(-\tfrac{1}{2}\right)|\uparrow\downarrow\rangle\right\} + |\downarrow\uparrow\rangle \neq ()|\uparrow\downarrow\rangle$$

similar for $|\downarrow\uparrow\rangle$

$= (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$

So S^2 matrix looks like

	$\|\uparrow\uparrow\rangle$	$\|\downarrow\downarrow\rangle$	$\|\uparrow\downarrow\rangle$	$\|\downarrow\uparrow\rangle$
$\langle\uparrow\uparrow\|$	2	0	0	0
$\langle\downarrow\downarrow\|$	0	2	0	0
$\langle\uparrow\downarrow\|$	0	0	1	1
$\langle\downarrow\uparrow\|$	0	0	1	1

BLOCK DIAGONAL!

$$S^2|\downarrow\uparrow\rangle = \underbrace{\left[\frac{3}{2} + 2\left(\tfrac{1}{2}\right)\left(-\tfrac{1}{2}\right)\right]}_{|\downarrow\uparrow\rangle}|\downarrow\uparrow\rangle + \underbrace{S_{1+}S_{2-}|\downarrow\uparrow\rangle}_{|\uparrow\downarrow\rangle} + S_{1-}S_{2+}|\downarrow\uparrow\rangle \to 0$$

We need the eigenvectors of $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ which are

obviously $e_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$ and $e_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$

e-values of S^2:

$\begin{bmatrix}1-\lambda & 1\\ 1 & 1-\lambda\end{bmatrix} = 0$

$\lambda = 0, 2$

$\to S(S+1)$

Thus the eigenfunctions of S^2 on the $|\uparrow\uparrow\rangle$ etc basis are

$|\uparrow\uparrow\rangle \quad (S=1)$

$|\downarrow\downarrow\rangle \quad (S=1)$

$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \quad (S=1)$

} Triplet $S = 1$

$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (S=0)$ } Singlet $S = 0$

The general problem

Consider $\vec{J} = \vec{J}_1 + \vec{J}_2$

The uncoupled representation is $|j_1\ j_2\ m_1\ m_2\rangle$

- j_1: eig of J_1^2
- j_2: eig of J_2^2
- m_1: eig of J_{1z}
- m_2: eig of J_{2z}

We want the "coupled" eigenvectors $|j_1\ j_2\ j\ m\rangle$

- j: eig of J^2
- m: eig of J_z

[Note, we can form simultaneous eigenstates of J^2, J_z, J_1^2, J_2^2 since they all commute with eachother]

Our problem is solved in general if we can write the coupled eigenvectors in terms of the uncoupled:

$$|j_1 j_2 j m\rangle = \sum_{\substack{j_1' j_2' \\ m_1 m_2}} |j_1' j_2' m_1 m_2\rangle \langle j_1' j_2' m_1 m_2 | j_1 j_2 j m\rangle$$

coupled — uncoupled — Clebsch-Gordan coefficient

General Rule for changing basis [representation].
The expansion coefficients for accomplishing this are the CG coefficients.

Many CG's vanish. Lets work out their properties

Note: $\langle j_1' j_2' m_1 m_2 | J_1^2 | j_1 j_2 j m \rangle = j_1(j_1+1) \langle j_1' j_2' m_1 m_2 | j_1 j_2 j m \rangle$

↳ or ↱

$= j_1'(j_1'+1) \langle j_1' j_2' m_1 m_2 | j_1 j_2 j m \rangle$

$\Rightarrow \langle 1 \rangle = 0$ unless $j_1 = j_1'$ (same for j_2)

Also:

$\langle j_1 j_2 m_1 m_2 | J_z | j_1 j_2 j m \rangle = (m_1+m_2) \langle j_1 j_2 m_1 m_2 | j_1 j_2 j m \rangle$

$= m \langle j_1 j_2 m_1 m_2 | j_1 j_2 j m \rangle$

$\Rightarrow \langle 1 \rangle = 0$ unless $m = m_1 + m_2$

So we can write:

$$|j_1 j_2 j m\rangle = \sum_{\substack{m_1 m_2 \\ (\text{such that } m_1+m_2=m)}} |j_1 j_2 m_1 m_2\rangle \langle j_1 j_2 m_1 m_2 | j_1 j_2 j m \rangle$$

At a practical level you can just look these up

General properties of CG's

① Orthogonality conditions

Since $|j_1 j_2 j m\rangle$ is a basis we have $|j_1 j_2 m_1 m_2\rangle = \sum_{\substack{j \\ m=m_1+m_2}} |j_1 j_2 j m\rangle\langle j_1 j_2 j m| j_1 j_2 m_1 m_2\rangle$

Hit this with $\langle j_1 j_2 m_1' m_2'|$, then

$$\langle j_1 j_2 m_1' m_2' | j_1 j_2 m_1 m_2\rangle = \sum_{jm} \langle j_1 j_2 m_1' m_2' | j_1 j_2 j m\rangle\langle j_1 j_2 j m | j_1 j_2 m_1 m_2\rangle$$

But LHS $\propto \delta_{m_1 m_1'}\, \delta_{m_2 m_2'}$ thus we have orthog. relation

$$\boxed{\sum_{jm} \langle j_1 j_2 m_1' m_2' | j_1 j_2 j m\rangle\langle j_1 j_2 j m | j_1 j_2 m_1 m_2\rangle = \delta_{m_1 m_1'}\, \delta_{m_2 m_2'}}$$

② Since CG's can be taken as real

$(CG)^* = (CG)$ or $\boxed{\langle j_1 j_2 j m | j_1 j_2 m_1 m_2\rangle = \langle j_1 j_2 m_1 m_2 | j_1 j_2 j m\rangle}$

③ Can also show that

$$\boxed{\sum_{m_1 m_2} \langle j_1 j_2 j m | j_1 j_2 m_1 m_2\rangle\langle j_1 j_2 m_1 m_2 | j_1 j_2 j' m'\rangle = \delta_{jj'}\,\delta_{mm'}}$$

④ Possible values of j (general):

$|j_1 - j_2| \le j \le j_1 + j_2$ triangle condition

ex: 2 spin $\frac{1}{2}$ $\quad 0 \le j \le 1$ $\quad$ eg $j = 0$ or 1

$j=0$: singlet; $j=1$: triplet

General formulae - easy to program. (Messiah Vol II Appendix C)

(1) Wigner 3j symbol

$$\begin{pmatrix} j_1 & j_2 & J \\ m_1 & m_2 & -M \end{pmatrix} \equiv \frac{(-)^{j_1 - j_2 + M}}{\sqrt{2J+1}} \langle j_1 j_2 m_1 m_2 | j_1 j_2 J M \rangle$$

(2) General Racah formula for 3j symbol

$$\begin{pmatrix} a & b & c \\ \alpha & \beta & \gamma \end{pmatrix} = (-)^{a-b-\gamma} \sqrt{\Delta} \sqrt{(a+\alpha)!(a-\alpha)!(b+\beta)!(b-\beta)!(c+\gamma)!(c-\gamma)!}$$

$$\times \sum_t (-)^t \left[t!(c-b+t+\alpha)!(c-a+t-\beta)!(a+b-c-t)!(a-t-\alpha)!(b-t+\beta)! \right]^{-1}$$

$(\alpha+\beta+\gamma = 0$, $|a-b| \le c \le a+b$ and

$$\Delta \equiv \frac{(a+b-c)!(b+c-a)!(c+a-b)!}{(a+b+c+1)!}$$

$\sum_t$ - over all t such that factorials have a meaning, eg argument of factorials are ≥ 0 $(0! = 1)$.

Messiah gives the most general commonly available discussion of CGs! Condon and Shortley has very detailed tables of CGC's and Wigner symbols. For three angular momenta and 6-j symbols see C&S.

Examples of CG coefficients

(ex-Schiff)

$j_1 = \frac{1}{2} \quad j_2 = \frac{1}{2}$:

	1	1	0	1	} j (total $\underset{\sim}{J}$)
	1	0	0	-1	} m
$\frac{1}{2}\ \frac{1}{2}$	1			0	
$\frac{1}{2}\ -\frac{1}{2}$		$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$		
$-\frac{1}{2}\ \frac{1}{2}$	0	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$	0	
$-\frac{1}{2}\ -\frac{1}{2}$		0		1	
$m_1\ m_2$					

Thus, to compute $\langle j_1 j_2 \tfrac{1}{2} -\tfrac{1}{2} | j_1 j_2 1 0 \rangle$ ($j_1 = j_2 = \frac{1}{2}$)

we get $+1/\sqrt{2}$ ($m_1\ m_2$; j, m)

To compute $\langle j_1 j_2 \tfrac{1}{2} \tfrac{1}{2} | j_1 j_2 1 -1 \rangle = 0$

etc

Tables for other j's

$\boxed{j_1 = 1 \quad j_2 = \frac{1}{2}}$:

m_1	m_2	m (total J)
1	$\frac{1}{2}$	$\frac{3}{2}$
1	$-\frac{1}{2}$	$\frac{1}{2}$
0	$\frac{1}{2}$	$\frac{1}{2}$
0	$-\frac{1}{2}$	$-\frac{1}{2}$
-1	$\frac{1}{2}$	$-\frac{1}{2}$
-1	$-\frac{1}{2}$	$-\frac{3}{2}$

m_1, m_2	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{3}{2}$
	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$
$1\ \frac{1}{2}$	1					
$1\ -\frac{1}{2}$		$\frac{1}{\sqrt{3}}$	$\sqrt{\frac{2}{3}}$			
$0\ \frac{1}{2}$		$\sqrt{\frac{2}{3}}$	$-\sqrt{\frac{1}{3}}$			
$0\ -\frac{1}{2}$				$\sqrt{\frac{2}{3}}$	$\sqrt{\frac{1}{3}}$	
$-1\ \frac{1}{2}$				$\sqrt{\frac{1}{3}}$	$-\sqrt{\frac{2}{3}}$	
$-1\ -\frac{1}{2}$						1

"Blank regions" of table are zero | j_m these parts $m \neq m_1 + m_2$.

$\boxed{j_1 = 1 \quad j_2 = 1}$:

m_1, m_2	2	2	1	2	1	0	2	1	2
	2	1	1	0	0	0	-1	-1	-2
$1\ 1$	1								
$1\ 0$		$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$						
$0\ 1$		$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$						
$1\ -1$				$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$			
$0\ 0$				$\sqrt{\frac{2}{3}}$	0	$-\sqrt{\frac{1}{3}}$			
$-1\ 1$				$\sqrt{\frac{1}{6}}$	$-\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$			
$0\ -1$							$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	
$-1\ 0$							$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$	
$-1\ -1$									1

General comments:

1) Can regard the aforementioned matrices as the transformation matrix from $|j_1 j_2 m_1 m_2\rangle$ representation into $|j_1 j_2 j m\rangle$ representation.

2) Note that the matrices are <u>block diagonal</u>. This means (from 1) that only uncoupled states $|j_1 j_2 m_1 m_2\rangle$ with $m_1 + m_2 = m$ are needed in the expansion for $|j_1 j_2 j m\rangle$.

<u>Question</u>: Would these matrices be block diagonal for arbitrary ordering of the rows & columns?

Explicit construction of CGC's

Recall that $|j_1-j_2| \le j \le j_1+j_2$ and $M = m_1+m_2$. This means that the coupled state:

$$|j_1, j_2, \underset{\downarrow}{j}{}_{j_1+j_2}, \underset{\downarrow}{m}{}_{m_1+m_2}\rangle = |j_1 j_2 \underset{\downarrow}{j_1}{}_{m_1} \underset{\downarrow}{j_2}{}_{m_2}\rangle \quad (*)$$

coupled represent. Product rep.

[The point is that for the case of $m_1 = j_1$ and $m_2 = j_2$ then the coupled eigenstate is exactly the product state]
This is useful for constructing CG's!

Let $J_\pm = J_x \pm i J_y = J_{1\pm} + J_{2\pm}$. Apply J_- to the "maximal state" $(*)$:

$$J_- |j_1 j_2 \overbrace{j_1+j_2}^{j} \overbrace{m_1+m_2}^{m}\rangle = (J_{1-} + J_{2-}) |j_1, j_2, j_1, j_2\rangle$$

$$J_- |jm\rangle = \sqrt{j(j+1) - m(m-1)}\, |j\, m-1\rangle$$

$$\Rightarrow$$

$$\sqrt{2(j_1+j_2)}\, |j_1 j_2\, j\, j-1\rangle = \sqrt{2j_1}\, |j_1 j_2\, j_1-1\, j_2\rangle + \sqrt{2j_2}\, |j_1\, j_2\, j_1\, j_2-1\rangle \quad (**)$$

close with the product bra $\langle j_1 j_2\, j_1-1\, j_2|$ to see that

$$\sqrt{2(j_1+j_2)}\, \langle j_1 j_2\, j_1-1\, j_2 | j_1 j_2\, j\, j-1\rangle = \sqrt{2j_1}\,(1) + \sqrt{2j_2}\, \langle j_1 j_2\, j_1-1\, j_2 | j_1 j_2\, j_1\, j_2-1\rangle$$

(other)

$$\Rightarrow$$

$$\langle j_1\, j_2\, j_1-1\, j_2 |\; j_1\, j_2\, j_1+j_2\, j_1+j_2-1\rangle = \sqrt{\frac{j_1}{j_1+j_2}}$$

Similar: $\langle j_1\, j_2\, j_1\, j_2-1 | j_1 j_2\, j_1+j_2\, j_1+j_2-1\rangle = \sqrt{\frac{j_2}{j_1+j_2}}$

Keep lowering until $|j_1 j_2\, j\, -j\rangle = |j_1\, j_2\, -j_1\, -j_2\rangle$ — manifold with total J_z reduced by 1!

Not done — $(**)$ must be $\perp$ to $|j_1 j_2\, j'\, j'\rangle$ for $j' = j_1+j_2-1$ thus we take

$$|j''j'\rangle = -\sqrt{\frac{j_2}{j_1+j_2}}\, |j_1 j_2\, j_1-1\, j_2\rangle + \sqrt{\frac{j_1}{j_1+j_2}}\, |j_1\, j_2\, j_1\, j_2-1\rangle \quad [\perp \text{ to } (**)]$$

Apply J_- again

Convention - choose phase so that coeff. of state with $|j_1, j_2, -1\rangle > 0$

↑ This is the "phase convention" (see Bett p 40). This determines that the right (with largest j_1) > 0. Then continue "lowering process".

Example: add $|1\alpha\rangle\ |\tfrac{1}{2}\beta\rangle$

Notation: $|j_1 j_2 | jm\rangle = ()\,|j_1 j_2\, m_1 m_2\rangle$ (LHS / RHS)

$$\left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{3}{2}\ \tfrac{3}{2}\right\rangle = \left|1\ \tfrac{1}{2}\ 1\ \tfrac{1}{2}\right\rangle$$

($j_1\ j_2\ j\ m$ on the left; $j_1\ j_2\ m_1\ m_2$ on the right)

maximum α, β

$J_-|jm\rangle = \sqrt{j(j+1) - m(m-1)}\,|j, m-1\rangle$

J_+ similar

Now lower: Apply J_- to both sides

$$\sqrt{3}\left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{3}{2}\ \tfrac{1}{2}\right\rangle = \sqrt{2}\left|1\ \tfrac{1}{2}\ 0\ \tfrac{1}{2}\right\rangle + 1\left|1\ \tfrac{1}{2}\ 1\ -\tfrac{1}{2}\right\rangle \quad (*)$$

$$\sqrt{3}\cdot 2\left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{3}{2}\ -\tfrac{1}{2}\right\rangle = \sqrt{2}\sqrt{2}\left|1\ \tfrac{1}{2}\ -1\ \tfrac{1}{2}\right\rangle + 2\sqrt{2}\left|1\ \tfrac{1}{2}\ 0\ -\tfrac{1}{2}\right\rangle$$

$$\sqrt{3}\sqrt{3}\cdot 2\left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{3}{2}\ -\tfrac{3}{2}\right\rangle = (4+2)\left|1\ \tfrac{1}{2}\ -1\ -\tfrac{1}{2}\right\rangle$$

$$\Rightarrow \left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{3}{2}\ \tfrac{3}{2}\right\rangle = \left|1\ \tfrac{1}{2}\ 1\ \tfrac{1}{2}\right\rangle$$

$$\Rightarrow \left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{3}{2}\ \tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\left|1\ \tfrac{1}{2}\ 0\ \tfrac{1}{2}\right\rangle + \sqrt{\tfrac{1}{3}}\left|1\ \tfrac{1}{2}\ 1\ -\tfrac{1}{2}\right\rangle$$

$$\Rightarrow \left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{3}{2}\ -\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{1}{3}}\left|1\ \tfrac{1}{2}\ -1\ \tfrac{1}{2}\right\rangle + \sqrt{\tfrac{2}{3}}\left|1\ \tfrac{1}{2}\ 0\ -\tfrac{1}{2}\right\rangle$$

$$\Rightarrow \left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{3}{2}\ -\tfrac{3}{2}\right\rangle = \left|1\ \tfrac{1}{2}\ -1\ -\tfrac{1}{2}\right\rangle$$

This gives all CG's for $|1\ \tfrac{1}{2}\ |\ \tfrac{3}{2}\ \alpha\rangle$ coupled states

What about $\left|1\ \tfrac{1}{2}\ \tfrac{1}{2}\ \tfrac{1}{2}\right\rangle$? ($j_1\ j_2\ j\ m$) Construct to be $\perp$ to $(*)$:

$$\left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{1}{2}\ \tfrac{1}{2}\right\rangle = -\frac{1}{\sqrt{3}}\left|1\ \tfrac{1}{2}\ 0\ \tfrac{1}{2}\right\rangle + \sqrt{\tfrac{2}{3}}\left|1\ \tfrac{1}{2}\ 1\ -\tfrac{1}{2}\right\rangle$$

(+ sign for piece with biggest $j_i > 0$)

$$\left|1\ \tfrac{1}{2}\ \middle|\ \tfrac{1}{2}\ -\tfrac{1}{2}\right\rangle = -\sqrt{\tfrac{2}{3}}\left|1\ \tfrac{1}{2}\ -1\ \tfrac{1}{2}\right\rangle + \sqrt{\tfrac{1}{3}}\left|1\ \tfrac{1}{2}\ 0\ -\tfrac{1}{2}\right\rangle$$

Gives CG's for $|1\ \tfrac{1}{2}\ |\ \tfrac{1}{2}\ \alpha\rangle$ states.

"Physics" example.

Spin-Orbit Coupling $|l\alpha\rangle|\tfrac{1}{2}\beta\rangle$ (orbital, spin)

$$\hat{H} = \hat{H}_0 + V(r)\,\vec{L}\cdot\vec{S}$$

$$\vec{L}\cdot\vec{S} = \tfrac{1}{2}(J^2 - L^2 - S^2)$$

Note: Could have done degenerate perturbation theory on product states — subspace matrix would have eigenvectors of total angular momentum!

Note that Hamiltonian is diagonal in total angular momentum states.

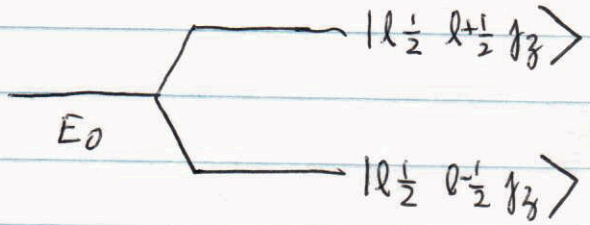
$$\Delta E = \langle \psi_0 | V(r)\, \underset{\sim}{L}\cdot\underset{\sim}{S} | \psi_0 \rangle$$

$$= \langle \psi_0(\vec{r}) | V(r) | \psi_0(\vec{r}) \rangle \langle l\tfrac{1}{2} j j_z | \tfrac{1}{2}(J^2 - L^2 - S^2) | l\tfrac{1}{2} j j_z \rangle$$

$$= \tfrac{1}{2} A\hbar^2 [j(j+1) - l(l+1) - s(s+1)]$$

For given l, $j = l \pm \tfrac{1}{2}$

$$\Delta E = \tfrac{1}{2}\hbar^2 A \begin{cases} l & \left(j = l + \tfrac{1}{2}\right) \\ -(l+1) & \left(j = l - \tfrac{1}{2}\right) \end{cases}$$



[Note: consider doing this in uncoupled representation, which is a bit more work]